

Semicon

“India’s Semiconductor Ecosystem May Surpass China Over Time”

Think gold is just for jewellery? In semiconductors, it determines whether chips last or fail. At SEMICON India '25, TANAKA GROUP leaders Yutaka Ito, Natalie Abe, and Satoshi Teshima shared with EFY’s Akanksha Sondhi Gaur and Nidhi Agarwal how ultra-pure bonding wires and recycled metals are making materials smarter, supply chains stronger, and positioning India as a true architect in the global semiconductor mission.



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What are ultra-high-purity materials, and why are they important in electronics?

When we talk about ultra-high-purity materials, we mean metals refined to an extraordinary level, often 99.99% pure or even higher, which experts call ‘four nines’ purity. This level of purity is critical for performance. Metals such as gold, silver, platinum, and palladium are essential for electronics to function properly. Even the tiniest impurities can create resistance, cause malfunctions, or shorten a device’s lifespan. By keeping materials ultra-pure, semiconductors and miniaturised devices can run smoothly, reliably, and efficiently for years. Producing such ultra-pure materials demands advanced technologies, including melting,

refining, and purity control, all areas in which TANAKA excels.

Why is the demand for these materials rising in advanced electronics?

Advanced packaging technologies, such as fine-pitch wire bonding and high-frequency chip integration, require clean surfaces and predictably conductive wires. Ultra-pure gold wires make this possible by allowing electrons to flow with minimal resistance.

For power electronics applications, thicker aluminium and copper wires or ribbons are employed. Their larger cross-section provides better current-carrying capacity and improved thermal robustness, which is essential for wide-bandgap devices such as silicon carbide (SiC) and gallium nitride (GaN).

In chiplet and 3D stacking technologies, interconnects increasingly depend on microbumps and hybrid bonding. Our silver adhesive paste enables effective die attachment and heat dissipation, maintaining reliability even under thermal stress. These approaches help meet the demands for high bandwidth, reliability, and compatibility with diverse substrates.

Have there been any breakthrough innovations in these materials in recent years?

Bonding wire technologies are evolving rapidly to meet next-generation applications. Aluminium ribbon bonding wires are now used to handle higher currents and power, particularly in electric vehicles and renewable energy systems. Copper wires, often coated